Asphaltene-Stabilized Emulsions, Chemical Demulsification, and the Role of Interfacial Aging

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April 5, 2026

1 The Problem of Asphaltene-Stabilized Emulsions

Crude oil produced from petroleum reservoirs commonly contains dispersed water in addition to its primary constituents, which are classified into saturates, aromatics, resins, and asphaltenes (SARA fractions). This water may originate from formation water naturally present in the reservoir, from water injected during secondary or tertiary recovery, or process water introduced during production and handling. Depending on crude composition, flow conditions, and water cut (the percentage of water in the produced stream), these aqueous phases can form persistent water-in-oil (W/O) emulsions that must be separated before the crude can be transported or refined. Emulsion stability is primarily controlled by components that adsorb at the oil-water interface, with asphaltenes playing the dominant stabilizing role.

Asphaltenes are operationally defined as the fraction of crude oil that is insoluble in low molecular weight alkanes such as n-heptane or n-pentane, but soluble in aromatic solvents such as toluene or xylene [1]. They typically consist of polycyclic aromatic hydrocarbon (PAH) cores containing four to ten fused rings, together with aliphatic side chains and heteroatoms (N, O, S). Individual asphaltene molecules generally have molecular weights in the range of 500–1000 Da and molecular diameter of approximately 1–2 nm. Within the colloidal description of crude oil, these molecules are present in a dispersed state stabilized by resins. When this dispersed state is destabilized by changes in pressure, temperature, or composition during production, the molecules begin to self-associate through π-π stacking between PAH cores, forming nanoaggregates of roughly 3–10 molecules with sizes of 3–5 nm. According to the supramolecular assembly model of Gray et al., hydrogen bonding, van der Waals forces, and electrostatic interactions also contribute to this hierarchical assembly. These nanoaggregates can further cluster into larger aggregates of 5–20 nm, which may eventually flocculate and deposit under sufficiently destabilizing conditions. Nanoparticles have been recently proposed as inhibitors of asphaltene aggregation, functioning by adsorbing on the nanoaggregate surface and further prevents cluster growth [15, 16].

A qualitatively similar hierarchy of association governs the behavior of asphaltenes at the oil-water interface, but with different consequences: instead of bulk deposition, the result is the formation of a mechanically resistant interfacial film that stabilizes water-in-oil emulsions [2, 14]. At low surface coverage, asphaltenes adsorb at the oil–water interface and typically lower the interfacial tension to some extent. As coverage increases, the interface develops measurable viscoelastic resistance, and mechanical stresses can become more important than changes in interfacial tension alone [5, 11, 19]. With continued adsorption and interfacial restructuring, adsorbed molecules or nanoaggregates may form laterally associated, heterogeneous interfacial structures through aromatic stacking and other intermolecular interactions [2, 5, 14]. This time-dependent evolution is commonly described as interfacial *aging* where the complex interfacial modulus $G^*(t)$ (a measure of how stiff the interface is) often increases by two to three orders of magnitude over timescales ranging from tens to thousands of seconds, reflecting a transition from a more viscous, mobile interface toward a more elastic and mechanically resistant one [3, 5]. However, the microscopic origin of this aging remains unresolved and may depend on the experimental setup, including continued adsorption from the bulk, heterogeneity and solvent quality [6]. Under conditions where the interfacial layer becomes strongly elastic or jammed, it can resist deformation under hydrodynamic forcing and suppress coalescence between droplets [5, 7, 8].

The resulting W/O emulsions are exceptionally stable and pose serious operational challenges throughout the petroleum production chain. Stable emulsions increase fluid viscosity and pumping cost, hinder phase separation, promote corrosion through retained water, and complicate downstream treatment and refining processes.

2 The Role of Chemical Demulsifiers

Breaking asphaltene-stabilized W/O emulsions is achieved through *demulsification*: the disruption of the protective interfacial film so that water droplets can coalesce and settle under gravity or centrifugal force. Chemical demulsification is the dominant industrial approach because it achieves orders-of-magnitude reductions in emulsion stability compared to other methods.

Mechanistically, an effective chemical demulsifier must accomplish three coupled tasks [4, 9]: (i) competitively adsorb at the oil-water interface and displace adsorbed asphaltene molecules; (ii) lower the interfacial elastic modulus G' so that the film can deform and rupture under the hydrodynamic pressure of a draining thin film; and (iii) promote droplet flocculation and coalescence, without itself forming a secondary stabilizing steric layer at excess concentrations [4, 6]. Of these three, criterion (ii) is often the most demanding for aged asphaltene films, because interfacial aging generally increases the elastic contribution to the rheological response and the resulting viscoelastic layer is more difficult to deform and rupture during thin-film drainage [3, 5, 7].

At low concentrations, such demulsifiers adsorb at the oil–water interface and reduce interfacial film rigidity through structural disruption of the adsorbed asphaltene layer. At higher concentrations, however, they may end up strengthening the interfacial film, reversing their intended demulsification effect.

2.1 Interfacial Rheology as a Design Criterion

Because the barrier to coalescence is the mechanical rigidity of the asphaltene film, interfacial rheology needs to be considered as a demulsifier screening criterion. A demulsifier that reduces IFT without lowering $G^*(\omega)$ will not destabilize a highly aged emulsion. Demulsifier adsorption and film softening act at the level of individual droplet interfaces, whereas thin film drainage occurs later, once droplets have collided and a thin aqueous bridge has formed between them. At that subsequent moment, the mechanical rigidity of the interface after demulsifier treatment determines whether the bridging film can be deformed and ruptured to complete coalescence [7].

Current limitations. Despite decades of study, some fundamental gaps remain in the understanding of asphaltene-stabilized emulsions. First, most experimental and computational studies have been conducted under near-ambient conditions, even though asphaltene adsorption, aggregation, and early interfacial aging may begin under the elevated temperature and pressure conditions encountered in reservoirs and production flowlines. Temperature and pressure alter asphaltene solubility, the kinetic barriers for adsorption and desorption, and the properties of the aqueous phase. No existing study maps how interfacial film structure and desorption kinetics depend on these variables at true high pressure, high temperature conditions. Second, no computational study(To the best of my knowledge) has characterized the *internal structural state* of the interfacial film and connect it to the desorption rate and the distribution of coalescence outcomes observed experimentally. Finally, most studies consider the role of asphaltene in model systems, while rarely composition of crude oil and the role of other components are studied.

3 Molecular Simulation of Asphaltene Systems

3.1 Molecular Dynamics: What Has Been Done

Atomistic MD simulation has been widely applied to asphaltene systems at the oil-water interface [9, 11, 13, 14, 18, 19]. Some results that have emerged:

- At low surface coverage, PAH cores orient predominantly perpendicular to the interface, transitioning to a disordered arrangement as asphaltene–asphaltene interactions become dominant at higher coverage [19].
- Umbrella sampling confirms that the aggregated interfacial state is the free-energy minimum, with π-π stacking and hydrogen bonding through polar groups providing the thermodynamic driving force for adsorption [19].
- At the level of film mechanics, Kobayashi [14] computed the dilatational modulus directly from MD trajectories, showing that dilatational modulus is governed by the most interfacially active molecular subcomponents and increases monotonically with surface concentration.
- In the context of demulsification, van der Waals interactions between demulsifier and asphaltene PAH cores are the primary driver of competitive interfacial displacement [9].

Force fields commonly used include GAFF, OPLS-AA, COMPASS, CVFF, and ReaxFF [13]. To the best of my knowledge, no machine learning potential parameterization yet exists for asphaltene-water interfaces.

3.2 Dissipative Particle Dynamics: What Has Been Done

Dissipative particle dynamics (DPD) operates at the mesoscale, representing groups of atoms as single beads whose conservative repulsion parameters a_{ij} are derived using atomistic MD simulations [12, 20]. This means the DPD model inherits the structural state of the atomistic reference system, which in all existing studies is a single equilibrium configuration. Song et al. [20] applied DPD to ultra heavy crude oil W/O emulsions and showed that which architecture of polyether demulsifiers controls both the final demulsification level and the early dehydration rate, and how coalescence proceeds as the simulation proceeds. This demonstrates that DPD can resolve mesoscale coalescence kinetics and rank demulsifier architectures, but in every case the interfacial film is initialized from a single equilibrium asphaltene configuration. Our DPD study will consider the structural state of the film prior to introducing the demulsifier.

3.3 The Aging Problem: Experimental Evidence That Simulation Has Not Addressed

Film evolution. Chang and Squires [5] measured the complex interfacial shear modulus $G^*(\omega, t)$ of asphaltene layers at the oil-water interface over aging times up to 6000 s using a double-wall ring (DWR) rheometer and a microbutton microrheometer with fluorescence visualization [5]. In seven nominally identical preparations, $G^*(t)$ increased by two to three orders of magnitude and showed no sign of approaching a plateau by the end of the measurement window [5]. The seven trials fell into three qualitatively distinct groups (low, intermediate, and high G^*) despite identical bulk asphaltene concentration, solvent composition, and protocol, indicating that some uncontrolled internal variable governs the aging trajectory [5].

Mullins et al. [3] confirmed the character of asphaltene adsorption across a range of brine conditions using oscillatory interfacial shear rheology. A two stage aging mechanism was identified [3]: Stage 1 is dominated by diffusion of small, mobile molecules (resins, low-MW aromatics) to the interface, during which the interface remains viscosity dominated; Stage 2 involves progressive formation of the asphaltene π-π contact network, during which G' grows faster than G'', driving the interface toward solid-like behavior.

Aging controls coalescence outcome. Chang [6] performed a systematic drop coalescence experiment in which a water droplet aged at the oil-water interface for a controlled time t_{age} was brought into contact with a flat water-oil interface and its fate recorded. Over 120 independent measurements:

- At $t_{\text{age}} = 80$ s: approximately **85%** of droplets coalesced within 200 s of contact; **0%** survived beyond 2200 s.
- At $t_{\text{age}} = 1080$ s: approximately **20%** of droplets coalesced within 200 s, while roughly **20%** survived indefinitely.

These results establish that the mechanical state of the asphaltene film at the time of contact, not the bulk asphaltene concentration, which was identical across all experiments determines whether coalescence occurs [6]. The transition shows no spontaneous recovery of coalescence capacity is observed once the film has aged [6, 8]. Furthermore, coalescence probability measured with microfluidic methods follows the same decrease with aging time [17].

The central mechanistic gap. The three distinct $G^*(t)$ groups observed by Chang and Squires [5] from *identical* preparations cannot be explained by differences in surface concentration Γ (molecules per unit area) alone, because Γ is set by the same bulk concentration and adsorption isotherm.

The central open question is therefore: what is the relationship between the internal structural state of the asphaltene film, its π-π contact network density and the desorption kinetics and coalescence outcomes that the film produces? Can we connect the macroscopic coalescence outcomes to the structural state of the film?

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